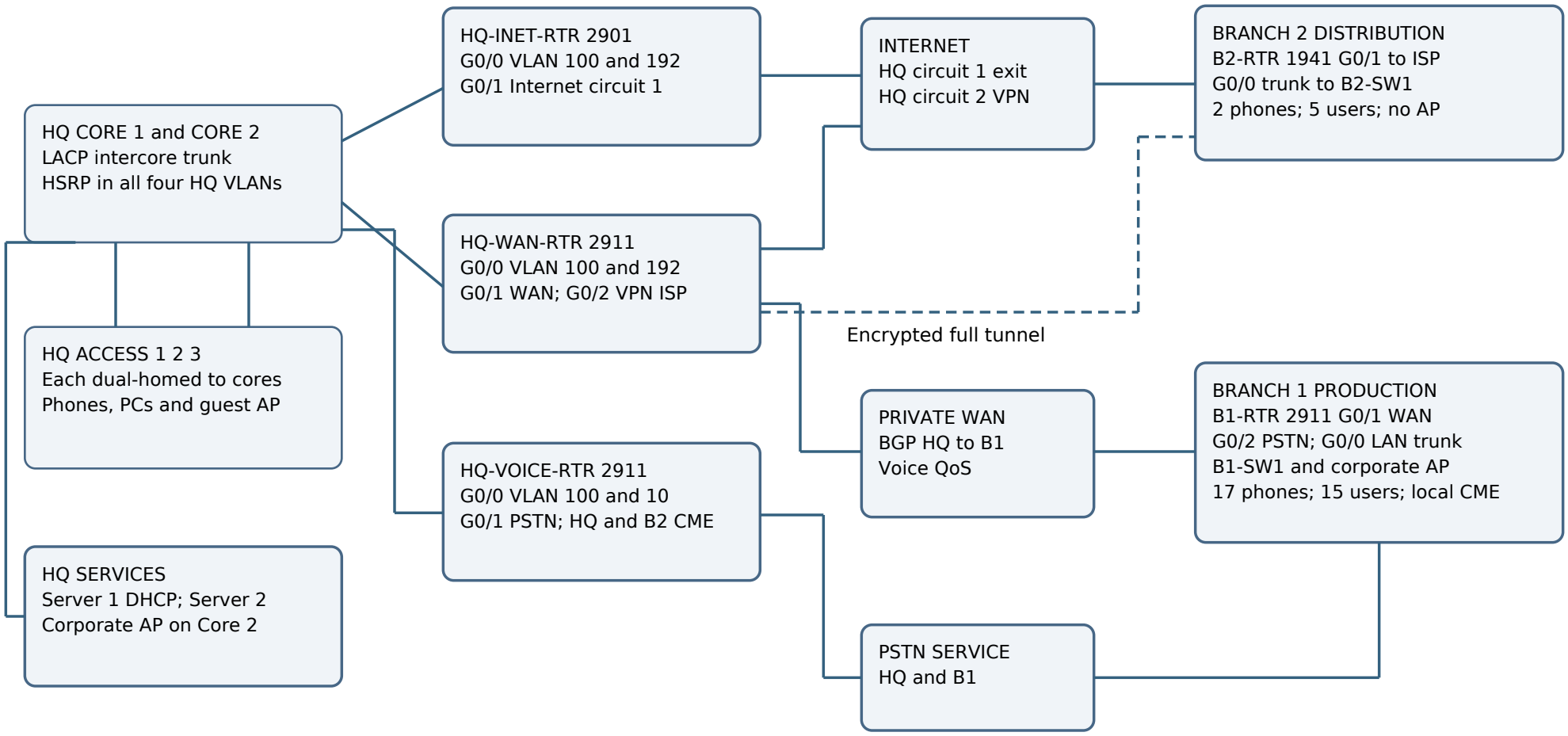
Ozark Precision Manufacturing Network Topology

Fictional company | Cody Phillips, Network Designer | Planned Packet Tracer build

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HQ users: 35 LAN users and 40 phones. Representative lab: HQ 3 phones, B1 2, B2 2.

B2 internet: B2 -> encrypted VPN -> HQ WAN -> HQ Internet NAT -> HQ circuit 1.

B1 internet: B1 -> private WAN -> HQ WAN -> HQ Internet NAT -> HQ circuit 1.

Solid lines show logical links. Dashed line shows VPN overlay. Provider details are pending.

Core redundancy does not provide redundant routers, servers or automatic ISP failover.

Network addressing and interface reference

VLAN	HQ subnet and VIP	B1 subnet and gateway	B2 subnet and gateway
100 Mgmt	192.168.110.0/24; .100	192.168.120.0/24; .1	192.168.130.0/24; .1
192 Data	192.168.10.0/24; .100	192.168.20.0/24; .1	192.168.30.0/24; .1
10 Voice	10.10.10.0/24; .100	10.10.20.0/24; .1	10.10.30.0/24; .1
172 Guest	172.16.10.0/24; .100	Not used	Not used

Device	LAN addresses and interfaces	External interfaces
HQ Core 1 / Core 2	SVIs .1 / .2 in each HQ VLAN; HSRP .100	Intercore LACP; dual access trunks
HQ-INET-RTR	G0/0.100 192.168.110.253; G0/0.192 192.168.10.253	G0/1 ISP 1
HQ-WAN-RTR	G0/0.100 192.168.110.254; G0/0.192 192.168.10.254	G0/1 private WAN; G0/2 ISP 2
HQ-VOICE-RTR	G0/0.100 192.168.110.252; G0/0.10 10.10.10.10	G0/1 PSTN
B1-RTR	G0/0.100 192.168.120.1; .192 192.168.20.1; .10 10.10.20.1	G0/1 WAN; G0/2 PSTN
B2-RTR	G0/0.100 192.168.130.1; .192 192.168.30.1; .10 10.10.30.1	G0/1 ISP and VPN

HQ access management: 192.168.110.241-.243. Branch switches: B1 192.168.120.254 and B2 192.168.130.254. Server 1: 192.168.10.50; Server 2 reserved: .51. APs proposed: HQ corporate 192.168.10.60; guest 172.16.10.60; B1 corporate 192.168.20.60.

HSRP/PVST primary: Core 1 VLANs 10 and 192; Core 2 VLANs 100 and 172. All masks /24. Phone access ports use data VLAN 192 and voice VLAN 10. Router subinterface numbers match VLAN tags. Actual physical switch port numbers require the starter file or a finalized cable map.

Example underlay and traffic paths

These addresses are proposed documentation-only lab values. They are not provider allocations. Replace the whole example consistently with the Packet Tracer starter-file or actual handoff plan. Do not mix either plan with the other.

Link	Customer address	Provider address
HQ Internet circuit 1	HQ-INET G0/1 203.0.113.2/30	ISP gateway 203.0.113.1
HQ VPN Internet circuit 2	HQ-WAN G0/2 198.51.100.2/30	ISP gateway 198.51.100.1
B2 Internet circuit	B2 G0/1 192.0.2.2/30	ISP gateway 192.0.2.1
HQ private WAN	HQ-WAN G0/1 10.255.0.2/30	WAN provider 10.255.0.1
B1 private WAN	B1 G0/1 10.255.0.6/30	WAN provider 10.255.0.5
HQ PSTN IP handoff	HQ-VOICE G0/1 10.255.1.2/30	PSTN peer 10.255.1.1
B1 PSTN IP handoff	B1 G0/2 10.255.1.6/30	PSTN peer 10.255.1.5

Example WAN provider ASN: 65000. Customer ASN: 65123 at HQ and B1; same-AS remedy required. The lab provider must route each public /30 and both WAN handoffs. It must carry B1 default traffic toward HQ and support the agreed customer prefix policy. These example provider devices are not part of the fictional customer bill of materials.

VPN peers in this example: HQ 198.51.100.2 and B2 192.0.2.2. B2 default uses 192.0.2.1; HQ WAN peer /32 and B2 /24 routes use 198.51.100.1. HQ WAN default uses HQ-INET data address 192.168.10.253. HQ-INET branch return routes use HQ-WAN data address 192.168.10.254.

HQ voice and guest destinations are reached by the Internet and WAN routers through 192.168.10.100. Management addresses appearing in the original diagram are for administration; do not use them in place of this data transit plan.

End-to-end voice media, full tunnel public destination forwarding, stateful filtering and same-AS BGP behavior require simulator evidence. No routing or security test is certified by this diagram.